



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

Mid-Semester Examinations, Sep-2026

Course Name: **Mechanics of Machinery**

Date: **21-09-2026**

Duration: **1.30 Hours**

Course Code: **ME203**

Time: **10:00 AM- 11:30 AM**

Max. Marks: **50**

ANSWER ALL QUESTIONS

Instructions: Answer all the questions (To-the-point). If additional data is required, assume it suitably and indicate the assumption in your solution. Use of AI will result in zero marks. A single A4 cheat sheet is attached. The figures may not be in proximity to the question. Use your time judiciously

1.

(a) Determine the mobility of the mechanisms shown in Fig. 1(a)-(d). Number each link and label the lower pairs and the higher pairs. (6)

(b) A planar mechanism is required to have 1 degree of freedom (DOF). The mechanism may contain binary, ternary, quaternary, pentagonal, and hexagonal links. If exactly two ternary links must be used and the maximum number of links to be used is 10, determine all possible combinations of the different types of links that satisfy the 1-DOF requirement. (4)

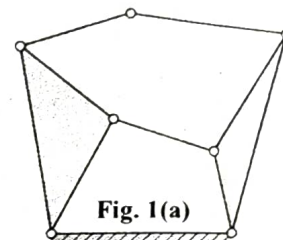
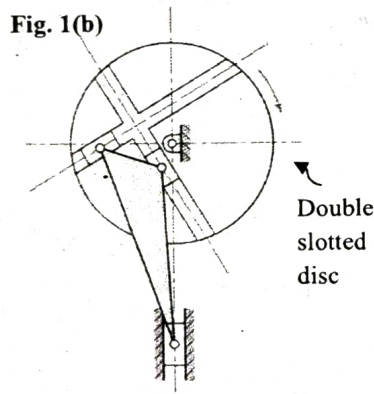
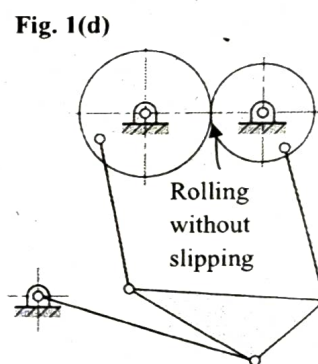
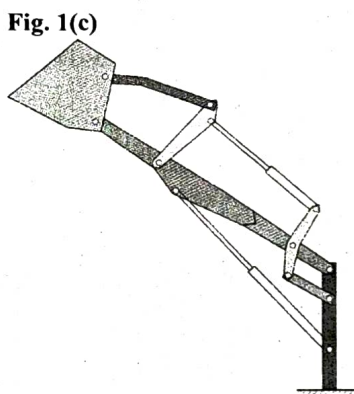


Fig. 1(a)



Double
slotted
disc



Rolling
without
slipping

2.

(a) Indicate whether the statements presented is true or false. If true, state why. If false, rewrite the statement to make it true. (3)

- "Any higher pair can be replaced by a lower pair, but this will increase the DOF by one."
- "Complete shrinkage of a higher-order link is equivalent to its removal." A multiple joint will be created, and the DOF will be reduced"

(b) For the four-bar mechanism shown (Fig. 2(a)), three link lengths are given as $r_1 = 1.0\text{cm}$, $r_3 = 2.5\text{cm}$, and $r_4 = 2.0\text{cm}$, while the length r_2 is variable. Determine the complete range of r_2 for which a crank-rocker mechanism can be obtained. Since any link may be fixed, identify the appropriate fixed link(s) for each valid range of r_2 . (4)

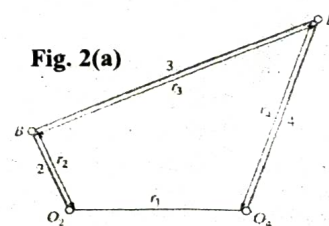
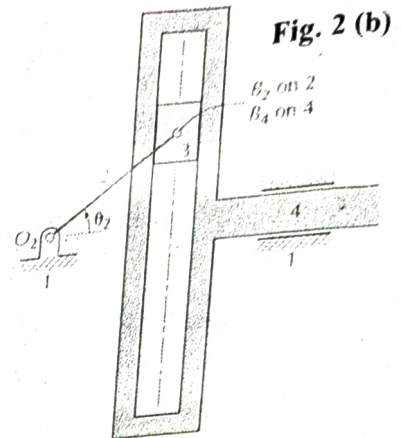


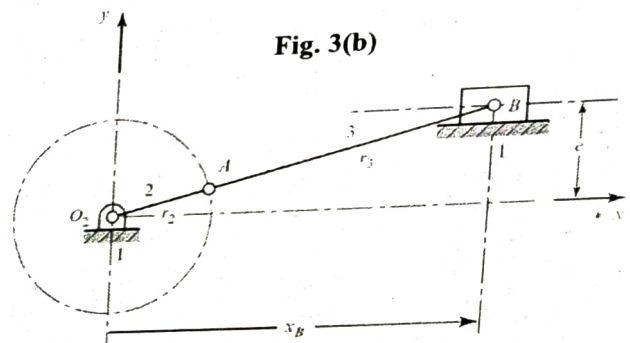
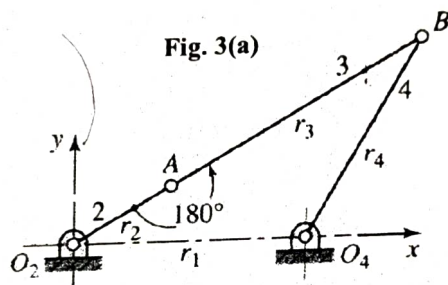
Fig. 2(a)

- (c) Discuss the different inversions of the Scotch yoke mechanism (Fig. 2(b)), clearly explaining the motion obtained in each inversion. (3)



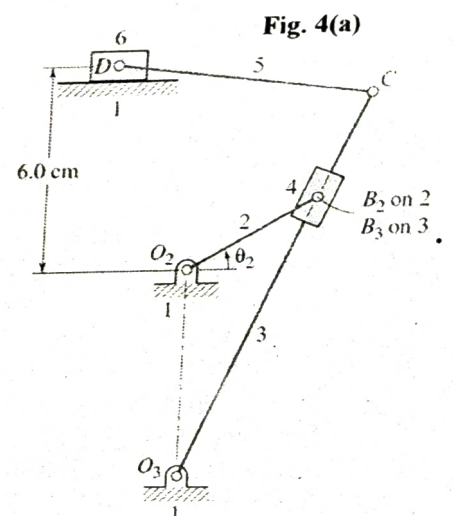
3. (a) Fig. 3(a) illustrates a crank and rocker linkage in the first of its two limit positions. In a limit position points O_2 , A, and B lie on a straight line; that is links 2 and 3 for a straight line. The two limit positions of a crank-rocker describe the extreme positions of the rocking angle. Suppose that such a linkage has $r_1 = 400\text{mm}$, $r_2 = 200\text{mm}$, $r_3 = 500\text{mm}$, and $r_4 = 400\text{mm}$. (i) Find θ_2 and θ_4 corresponding to each limit position. (ii) What is the total rocking angle of link 4. (iii) What are the transmission angles at the extremes? (5)

- (b) The Fig. 3(b) shows a slider crank mechanism that has an offset e and that is placed in one of its limiting positions. By changing the offset e , it is possible to cause the angle that crank 2 makes in traversing between the two limiting positions to vary in such a manner that the driving of forward stroke of the slider takes place over a large angle than the angle used for the return stroke. Such a linkage is then called a quick-return mechanism. The problem here is to find (i) the time ratio of the drive to return strokes. (ii) Also determine which direction the crank should rotate. (iii) Where do you think the velocity of the slider will be maximum. (3)



4. (a) In a four-bar chain ABCD, AB is the driving link, CD the driven link and AD the fixed link. Show that the angular velocity of CD is to that of AB as QA is to QD, where Q is the point of intersection of BC and AD, produced if necessary. When the links AB, BC, CD and DA are respectively 250 mm, 700mm, 450 mm and 800 mm. long, the angle BAD is 60° , AB and DC are on opposite sides of AD and the velocity of B is 3 m/s., find the velocity of C and the angular velocity of BC. (4).

- (b) For the mechanism shown in Fig. 4(a), point out all the instantaneous centers and determine the linear velocity of link 6. Given that $r_{O_2O_3} = 6.0\text{cm}$; $r_{O_2B_2} = 4.0\text{cm}$; $r_{O_3C} = 11.0\text{cm}$; $r_{CD} = 5.0\text{cm}$; $\theta_2 = 30^\circ$; $\dot{\theta}_2 = 20\text{ rad/sec (CCW)}$. (6)



- (c) The dimensions of the *Andreau differential-stroke-engine* mechanism, Fig. 4(b), are: $OA = 73 \text{ mm}$, $QB = 36.5 \text{ mm}$, $AC = BC = 150 \text{ mm}$, $CP = 214 \text{ mm}$. OA and QB are geared together so that QB turns at twice the speed of OA and in the opposite sense to OA . Find the velocity of the piston P for the given configuration, when OA makes 700 r.p.m. (3.5). (O and Q are fixed)

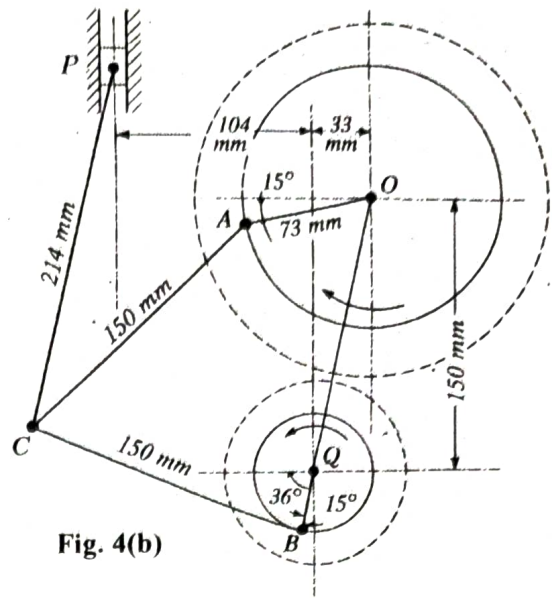


Fig. 4(b)

5.
(a) Indicate whether the statement presented is true or false. If true, state why. If false, rewrite the statement to make it true. "For a mechanism shown in Fig. 5(a), the mechanical advantage for the given configuration is zero" (1)



Fig. 5(a)

- (b) A slider moving outward on a rod OR (Fig. 5(b)) with a velocity of 3 m/s and acceleration of 9 m/s^2 . The rod has an angular velocity of 7 rad/s counterclockwise about O and an angular acceleration of 12 rad/s^2 clockwise. Find the magnitude of absolute acceleration of the slider at location A in m/s^2 . (3)

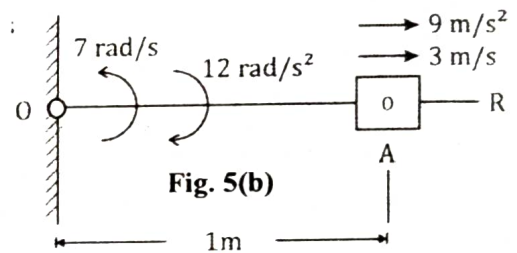


Fig. 5(b)

- (c) *Bennett's Construction*: Divide CP at R, such that $CR = CO^2/CP$. This may be done graphically by taking OC at right angles to the line of stroke of P and dropping the perpendicular OR on to the corresponding position of CP. For any other crank position, draw RS perpendicular to CP to meet the line of stroke at S, draw SQ perpendicular to the line of stroke to meet CP at Q and draw QN perpendicular to CP. Prove that Bennett's Construction results to the same acceleration diagram as Klein's construction. Klein's construction is shown in Fig. 5(c) (4.5).

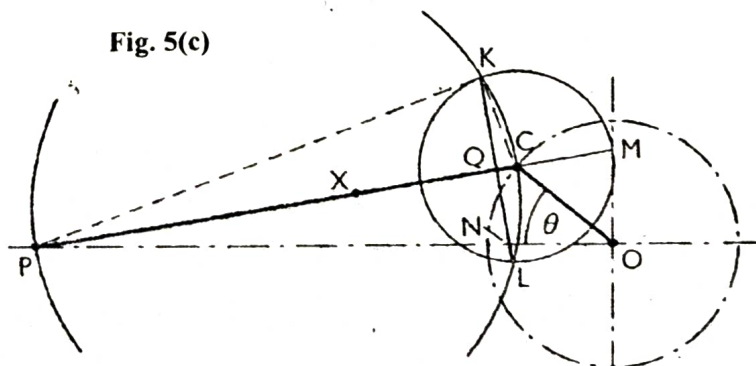


Fig. 5(c)